Adaptive Lane Detection Under Varying Lighting for Resource-Constrained Autonomous Systems

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1 Additional Qualitative Results

This section presents additional qualitative comparisons between the proposed method and the two baseline methods under different illumination conditions. Representative examples from the CULane test set are shown for daytime, shadow, and night-time scenarios to complement the quantitative results in the main paper. These examples further demonstrate the robustness of the proposed method in maintaining reliable lane detection across varying lighting conditions.

1.1 Daytime Detection Comparison

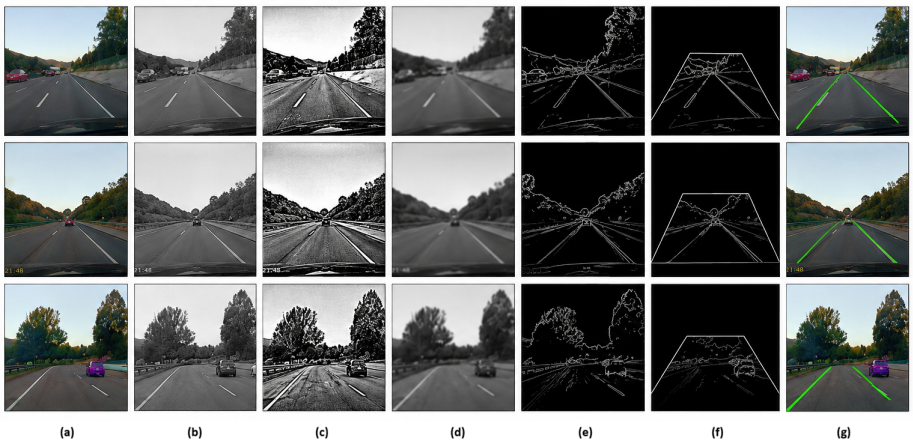


Fig. 1: Pipeline of Method 1 (Hough Transform) during daytime conditions on the CULane dataset. (a) Original input image. (b) Grayscale conversion. (c) CLAHE contrast enhancement. (d) Gaussian blur for noise reduction. (e) Canny edge detection. (f) Trapezoidal ROI masking (Top 20%–80% region). (g) Hough line detection with detected lane overlay (green line). The blue dashed line indicates the lane centre reference, and the black line represents the ROI boundary.

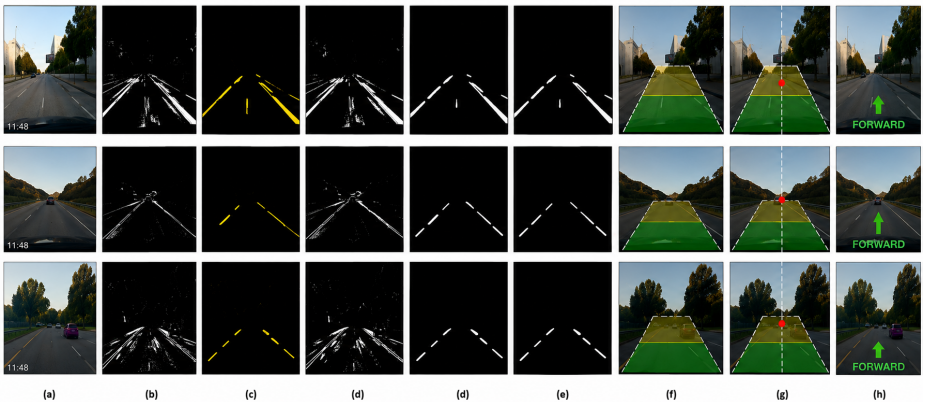


Fig. 2: Pipeline of Method 2 (Fixed HSV) under daytime conditions on the CULane dataset. (a) Input image. (b) HSV colour space conversion, (c) fixed HSV thresholding. (d) morphological filtering. (e) trapezoidal ROI masking. (f) two-zone segmentation. (g) weighted centroid estimation, and (h) final lane detection. The *green line* indicates the detected lane position, and the *red dashed line* denotes the boundary between the near and far zones.

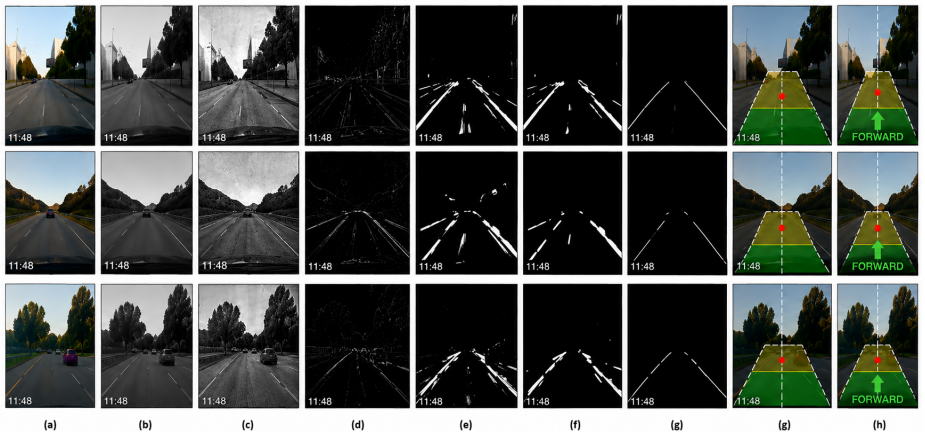


Fig. 3: Pipeline of Method 3 (Proposed Adaptive HSV) during daytime conditions on the CULane dataset. (a) Original input image. (b) HSV colour space conversion. (c) Adaptive HSV thresholding with brightness-aware thresholds. (d) Erosion and dilation for noise removal. (e) Trapezoidal ROI masking. (f) Two-zone segmentation (Near/Far zones). (g) Mode-based centroid calculation with histogram peak selection. (h) Final lane position overlay. The *green line* indicates the detected lane position, and the *red dashed line* shows the zone boundary.

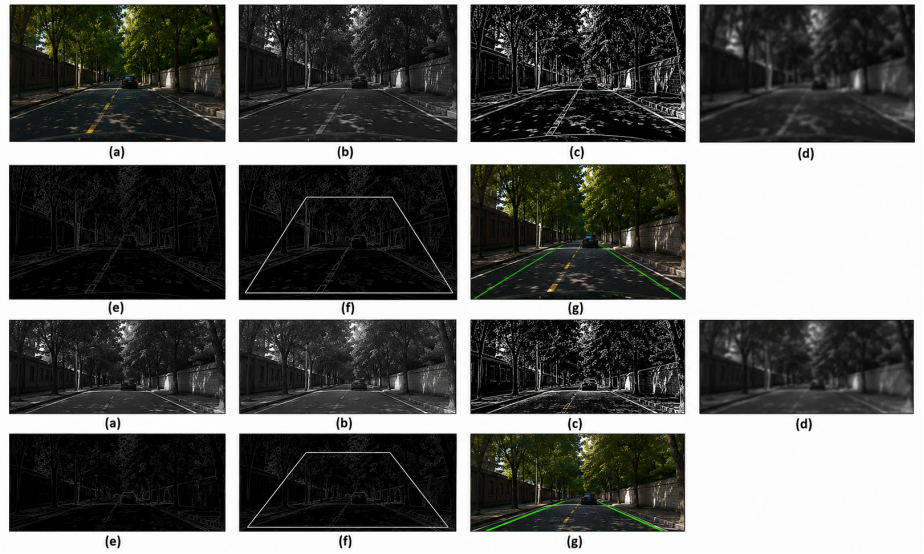


Fig. 4: Pipeline of Method 1 (Hough Transform) under shadow conditions on the CULane dataset. (a) Input image. (b) grayscale conversion, (c) CLAHE contrast enhancement. (d) Gaussian blur. (e) Canny edge detection. (f) trapezoidal ROI masking. and (g) Hough line detection with the final lane overlay. The green line indicates the detected lane position, the blue dashed line represents the lane centre reference, the black line denotes the ROI boundary, and the red arrows highlight fragmented detections at shadow boundaries.

Under normal daytime conditions, all three methods perform reasonably well. The Hough baseline produces fragmented detections where lane markings are partially occluded or worn, while the fixed HSV baseline occasionally misclassifies road texture as lane markings. The proposed method maintains clean and continuous lane boundaries through adaptive thresholding, morphological refinement, and two-zone weighted centroid estimation.

Quantitatively, the Hough baseline achieves the highest daytime accuracy (77.9%), followed by the fixed HSV baseline (72.7%) and the proposed method (71.2%). Although the proposed method shows a slight reduction in daytime accuracy, it provides substantially improved robustness under shadow and nighttime conditions while maintaining real-time performance at 686.8 FPS on a CPU-only platform.

1.2 Shadow Detection Comparison

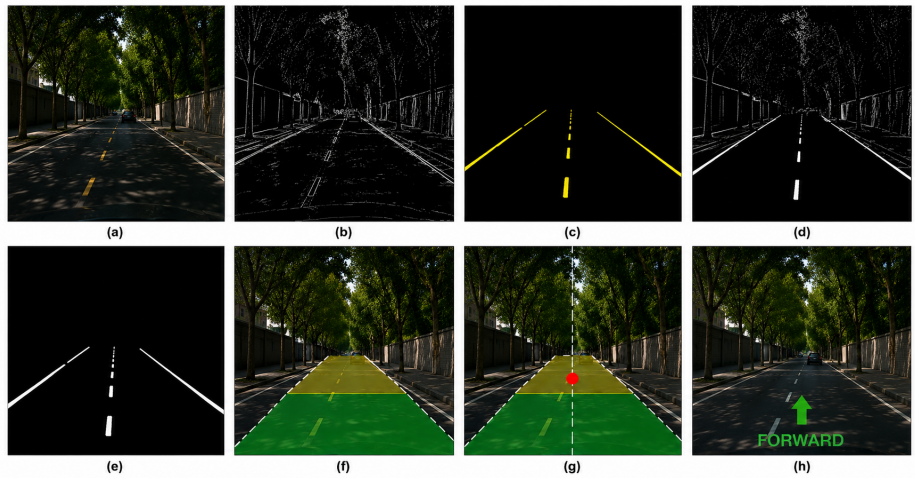


Fig. 5: Pipeline of Method 2 (Fixed HSV) under shadow conditions on the CU-Lane dataset. (a) Input image. (b) HSV colour space conversion. (c) fixed HSV thresholding. (d) morphological filtering. (e) trapezoidal ROI masking. (f) two-zone segmentation. (g) weighted centroid estimation and (h) final lane detection. The green line indicates the detected lane position, the blue dashed line represents the lane centre reference, the red dashed line denotes the boundary between the near and far zones, and the red arrows highlight lane markings missed by the fixed HSV threshold.

Under shadow conditions, the Hough baseline produces fragmented edges at shadow boundaries due to sudden contrast changes. The fixed HSV baseline misses lane pixels because its fixed Value threshold cannot compensate for reduced brightness. In contrast, the proposed method adapts the Value threshold to the local scene brightness, maintaining more continuous and reliable lane detection across shadow boundaries.

Quantitatively, the proposed method achieves 73.4% shadow accuracy, substantially outperforming the fixed HSV baseline (20.9%) and closely matching the Hough baseline (76.1%). This 52.5 percentage point improvement over the fixed HSV baseline demonstrates the effectiveness of adaptive thresholding under challenging illumination conditions.

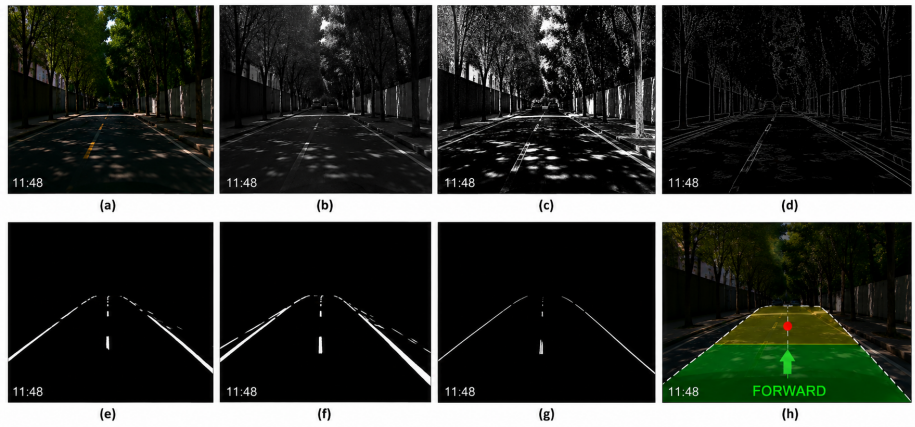


Fig. 6: Pipeline of Method 3 (Proposed Adaptive HSV) under shadow conditions on the CULane dataset. (a) Input image. (b) HSV colour space conversion. (c) adaptive HSV thresholding. (d) morphological filtering. (e) trapezoidal ROI masking. (f) two-zone segmentation. (g) mode-based centroid estimation and (h) final lane detection. The green line indicates the detected lane position, the blue dashed line represents the lane centre reference, the red dashed line denotes the boundary between the near and far zones, and the green arrows highlight lane markings successfully detected under shadow conditions.

1.3 Night time Detection Comparison

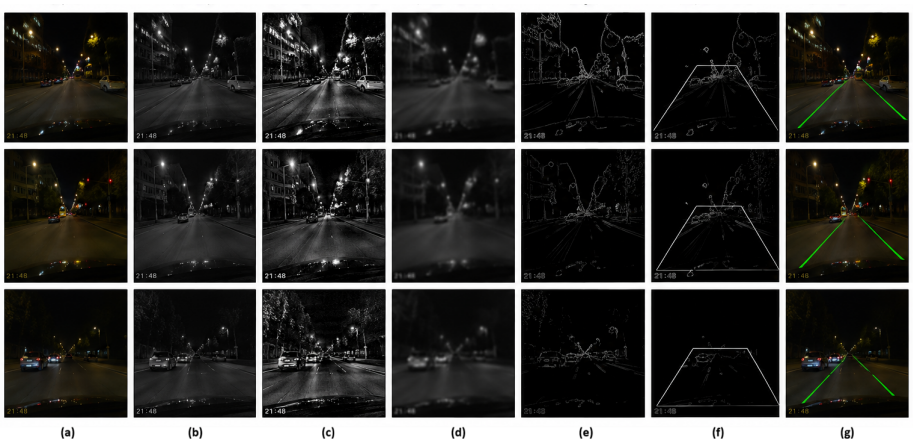


Fig. 7: Pipeline of Method 1 (Hough Transform) under night-time conditions on the CULane dataset. (a) Input image. (b) grayscale conversion. (c) CLAHE contrast enhancement. (d) Gaussian blur. (e) Canny edge detection. (f) trapezoidal ROI masking and (g) Hough line detection with the final lane overlay. The*blue dashed line* represents the lane centre reference, the black line denotes the ROI boundary, and the *red arrows* highlight regions where weak edge responses lead to missed lane detections.

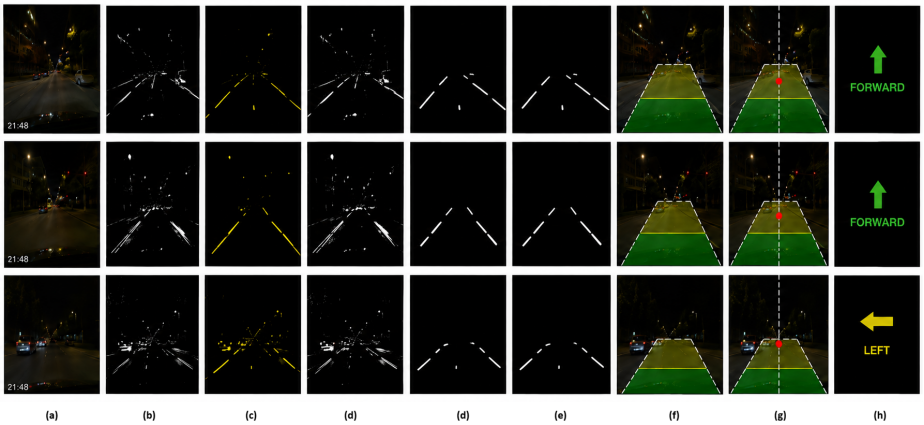


Fig. 8: Pipeline of Method 2 (Fixed HSV) under night-time conditions on the CULane dataset. (a) Input image. (b) HSV colour space conversion. (c) fixed HSV thresholding. (d) morphological filtering. (e) trapezoidal ROI masking, (f) two-zone segmentation. (g) weighted centroid estimation, and (h) final lane detection. The *green line* indicates the detected lane position, the *blue dashed line* represents the lane centre reference. the *red dashed line* denotes the boundary between the near and far zones, and the *red arrows* highlight lane markings missed because the fixed threshold excludes under-exposed lane pixels.

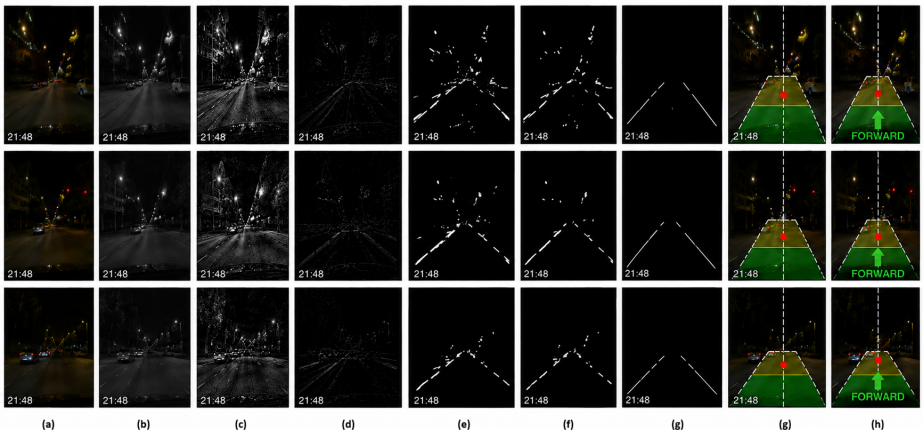


Fig. 9: Pipeline of Method 3 (Proposed Adaptive HSV) under night-time conditions on the CULane dataset. (a) Input image, (b) HSV colour space conversion, (c) Adaptive HSV thresholding using brightness-aware thresholds, (d) Morphological filtering, (e) Trapezoidal ROI masking, (f) Two-zone segmentation, (g) Mode-based centroid estimation, and (h) Final lane detection. The *green line* indicates the detected lane position, the *blue dashed line* represents the lane centre reference, and the *red dashed line* denotes the boundary between the near and far zones.

Under night-time conditions, the Hough baseline fails to detect lane markings due to weak edge responses, while the fixed HSV baseline produces sparse detections because its fixed threshold excludes many under-exposed lane pixels. In contrast, the proposed method combines CLAHE contrast enhancement, adaptive HSV thresholding, gamma correction, and Top-Hat filtering to produce more continuous and reliable lane detection.

Quantitatively, the proposed method achieves 70.8% night-time accuracy, outperforming both the Hough baseline (54.1%) and the fixed HSV baseline (36.9%). The largest performance gain is provided by adaptive HSV thresholding, with its removal reducing overall accuracy by 28.6 percentage points, highlighting its importance for robust low-light detection.

Overall, the Hough and fixed HSV baselines perform reasonably well under favourable daytime conditions but degrade significantly under shadows and night-time illumination. In contrast, the proposed method maintains more consistent performance across all lighting conditions, achieving only a modest reduction in daytime accuracy while providing substantial improvements under shadow and night-time scenarios.